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About

Prof. Yiu-ming Cheung received his PhD from the Department of Computer Science and Engineering at The Chinese University of Hong Kong. He is currently Chair Professor (Artificial Intelligence) in the Department of Computer Science, and Dean of the Institute for Research and Continuing Education (IRACE) at Hong Kong Baptist University (HKBU). Prof. Cheung is a Member of European Academy of Sciences and Arts (MEASA), and a Fellow of the IEEE, AAAS, IAPR, IET, and BCS. He is a recipient of 2023-2024 RGC Senior Research Fellow Award, the recipient of the HKBU President's Award 2023–2024 for Outstanding Performance in Scholarly Work, and the 2023 APNNS Outstanding Achievement Award, among others. Prof. Cheung was elected Distinguished Lecturer of the IEEE Computational Intelligence Society in 2020 and has been named a Changjiang Scholar (Chair Professor) by the Ministry of Education of China. He is currently serving as Editor-in-Chief of the IEEE Transactions on Emerging Topics in Computational Intelligence, and is serving / has served as Associate Editor for several leading journals, e.g., IEEE Transactions on Cognitive and Developmental Systems, ACM Transactions on Intelligent Systems and Technology, Pattern Recognition, and Neurocomputing.

Research Interests

- Machine Learning
- Visual Computing
- Data Science
- Pattern Recognition
- Multi-objective Optimization
- Information Security

Selected Publications

1. H.M. Cai, Y. Hu, F. Qi, B. Hu and **Y.M. Cheung***, “Deep Tensor Spectral Clustering Network via Ensemble of Multiple Affinity Tensors”, *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(7): 5080-5091, July 2024.
2. H.M. Cai, W.T. Huang, S.R. Yang, S.Q. Ding, Y. Zhang, B. Hu, F. Zhang and **Y.M. Cheung***, “Realize Generative yet Complete Latent Representation for Incomplete Multi-view Learning”, *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(5): 3637-3652, May 2024.
3. W.C. Lan, **Y.M. Cheung***, J.Y. Jiang, Z.K. Hu and M.K. Li, “Compact Neural Network via Stacking Hybrid Units”, *IEEE Transactions on Pattern Analysis and Machine Intelligence*, Vol. 46(1): 103-116, January 2024.

4. M.K. Li, **Y.M. Cheung*** and Z.K. Hu, “Key Point Sensitive Loss for Long-tailed Visual Recognition”, *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(4): 4812-4825, 2023.
5. Y.Q. Zhang and **Y.M. Cheung***, “Learnable Weighting of Intra-attribute Distances for Categorical Data Clustering with Nominal and Ordinal Attributes”, *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(7): 3560-3576, 2022.
6. X. Liu, K. Hu, H.B. Ling and **Y.M. Cheung***, “MTFH: A Matrix Tri-Factorization Hashing Framework for Efficient Cross-Modal Retrieval”, *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 43(3): 964-981, 2021.
7. Y. Zhou and **Y.M. Cheung***, “Bayesian Low-Tubal-Rank Robust Tensor Factorization with Multi-Rank Determination”, *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 43(1): 62-76, 2021.
8. **Y.M. Cheung*** and Y.Q. Zhang, “Fast and Accurate Hierarchical Clustering Based on Growing Multi-Layer Topology Training”, *IEEE Transactions on Neural Networks and Learning Systems*, 30(3): 876-890, 2019.
9. H. Zeng and **Y.M. Cheung***, “Feature Selection and Kernel Learning for Local Learning Based Clustering”, *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 33(8): 1532-1547, 2011.
10. **Y.M. Cheung***, “Maximum Weighted Likelihood via Rival Penalized EM for Density Mixture Clustering with Automatic Model Selection”, *IEEE Transactions on Knowledge and Data Engineering*, 17(6): 750-761, 2005.

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